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Section 11

① The Opcode field in a machine code instruction defines the specific operation that the processor must execute, such as addition, subtraction, data loading or branching. It is a critical part of the instruction because it tells the CPU what action to take. Depending on the opcode, the CPU knows how to interpret the remaining bits in the instruction. It also helps the control unit generate the right signals to activate components. Overall, the opcode determines the functionality of the instruction.

② In RISC-V, the `funct3` field specifies the size and type of data to be loaded into a register. For the instruction `LD X9, 10[X21]`, the machine uses the `funct3` field with value 011 to recognize it as a load doubleword (64-bit) operation. This field helps distinguish between loading bytes, halfwords, words or doublewords. Without `funct3`, the CPU wouldn't know how many bits to fetch from memory. It works alongside the opcode to fully define the instruction's behavior.

③ a)

A = X20

B = X21

g = X18

If 1:

ld x28, 24(X20)

ld x29, 48(X20)

Beq x28, x29, else1

If 2:

ld x5, 24(X20)

BNE x5, x0, else2

add x5, x5, x18

sd x5, 24(X20)

Beq x0, x0, exit

else2:

ld x5, 48(X20)

slli x5, x5, 4

sd x5, 48(X20)

Beq x0, x0, exit

else1:

ld x28, 48(X20)

slli x29, x28, 3

~~sd x28,~~

add x28, x28, x29

sd x28, 48(X20)

exit

b) $\text{ld } x_{28}, 24(x_{20}) \rightarrow \text{I type}$

0000	0001	1000	10100	XXX	11100	XXXXXXXX
			RS1	funct3	RD	Opcode

$\text{add } x_5, x_5, x_{18} \rightarrow \text{R type}$

XXXXXXXX	10010	00101	XXX	00101	XXXXXXXX
funct7	rs2	rs1	funct3	rd	Opcode

$\text{sd } x_5, 24(x_{20}) \rightarrow \text{S type}$

0000000	00101	10100	XXX	11000	XXXXXXXX
imm[11:5]	rs2	rs1	funct3	imm[4:0]	Opcode

④

$\text{ld } x_5, 40(x_{21})$

$\text{add } x_5, x_5, x_{18}$

$\text{slli } x_6, x_{22}, 3$

$\text{add } x_6, x_6, x_2$

$\text{ld } x_7, 0(x_6)$

$\text{slli } x_7, x_7, 3$

$\text{add } x_7, x_7, x_{20}$

$\text{sd } x_5, 0(x_7)$

$A = x_{20}$

$B = x_{21}$

$g = x_{18}$

⑤ The funct3 field in a load instruction in RISC-V specifies the type and size of data to be loaded. It tells the processor whether to load a byte, halfword, word or doubleword and whether the data should be signed or unsigned. This allows the same LOAD opcode to handle multiple data formats. This field is essential for correctly interpreting how much data to read from memory and how to handle it in registers.

①

(15X) 0P 2Y 1L
 21Y 2Y 2Y 11b
 31Y 3Y 3Y 11b
 15Y 3Y 3Y 11b
 31Y 3Y 3Y 11b
 31Y 3Y 3Y 11b
 31Y 3Y 3Y 11b
 31Y 3Y 3Y 11b
 31Y 3Y 3Y 11b